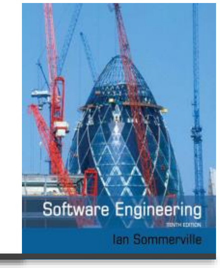
Chapter 2 – Software Processes

Topics covered



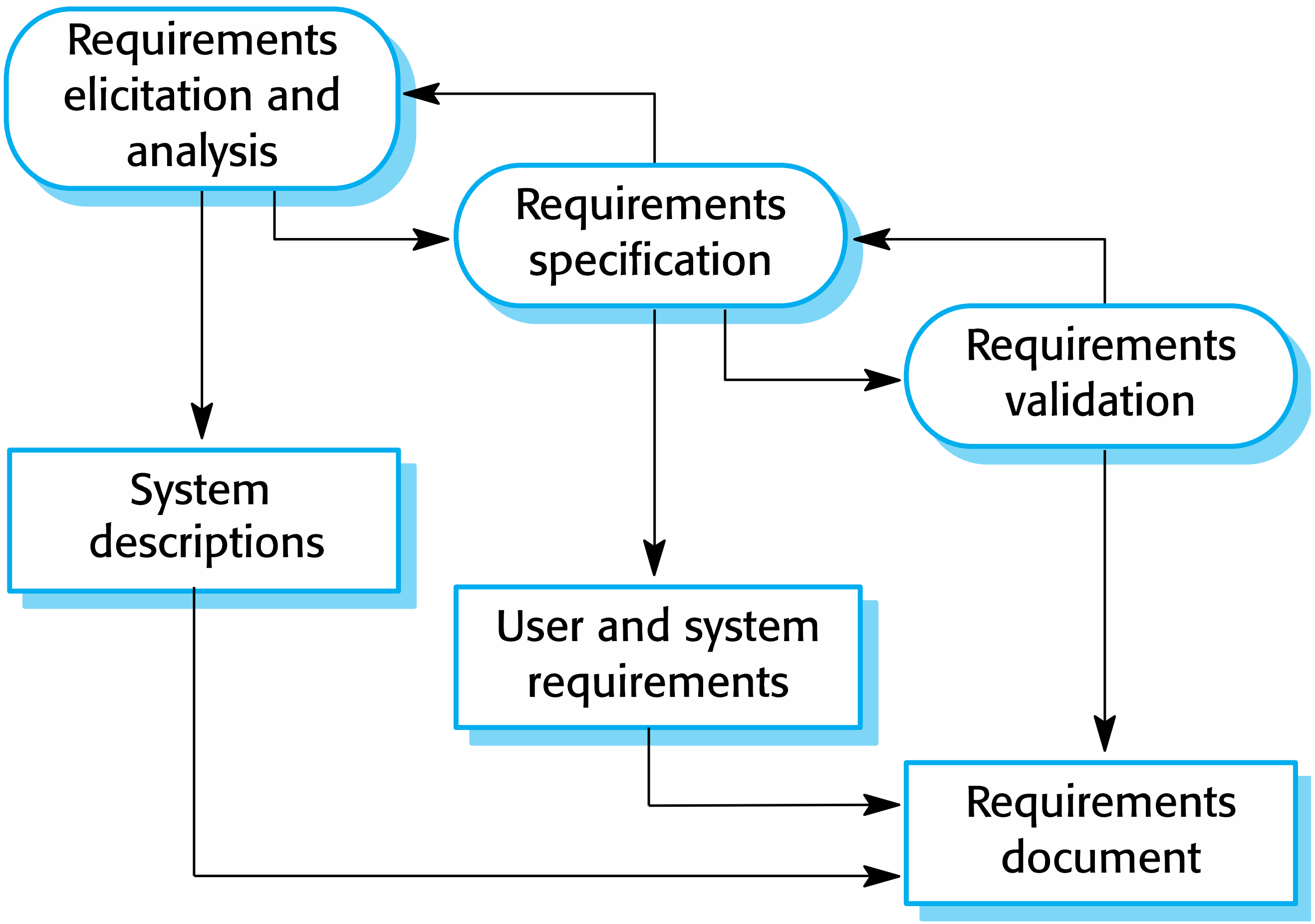
- Process activities
- Software process models

Process activities

Process activities

- The four basic process activities of specification, development, validation and evolution are organized differently in different development processes.

The requirements engineering process



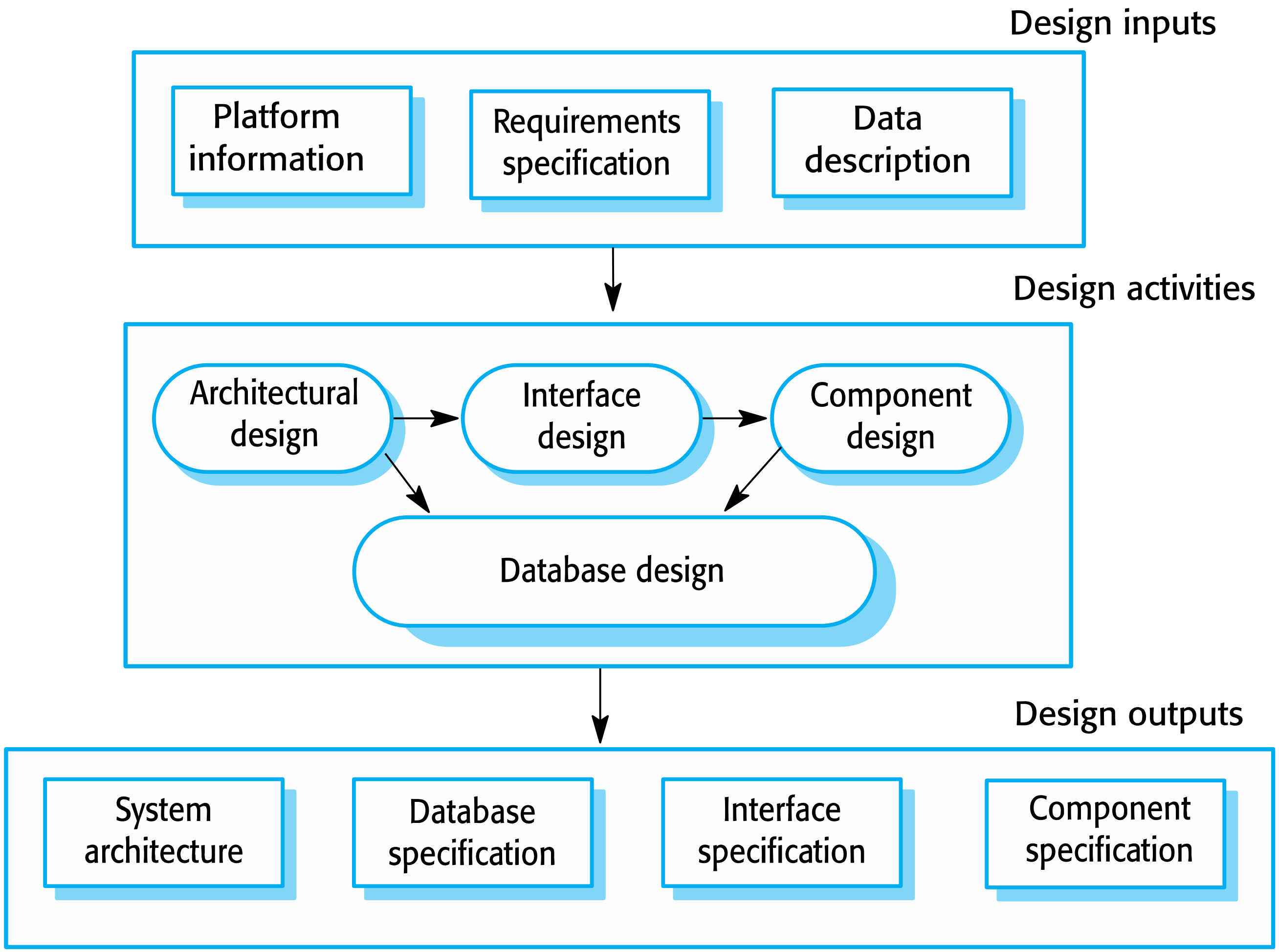
Software specification

- ✧ The process of establishing what services are required and the constraints on the system's operation and development.
- ✧ Requirements engineering process
  - Requirements elicitation and analysis
    - What do the system stakeholders require or expect from the system?
  - Requirements specification
    - Defining the requirements in detail
  - Requirements validation
    - Checking the validity of the requirements

Software design and implementation

- ✧ The process of converting the system specification into an executable system.
- ✧ Software design
  - Design a software structure that realises the specification;
- ✧ Implementation
  - Translate this structure into an executable program;
- ✧ The activities of design and implementation are closely related and may be inter-leaved.

A general model of the design process



Design activities

- *Architectural design,* where you identify the overall structure of the system, the principal components (subsystems or modules), their relationships and how they are distributed.
- *Database design,* where you design the system data structures and how these are to be represented in a database.
- *Interface design,* where you define the interfaces between system components.
- *Component selection and design,* where you search for reusable components. If unavailable, you design how it will operate.

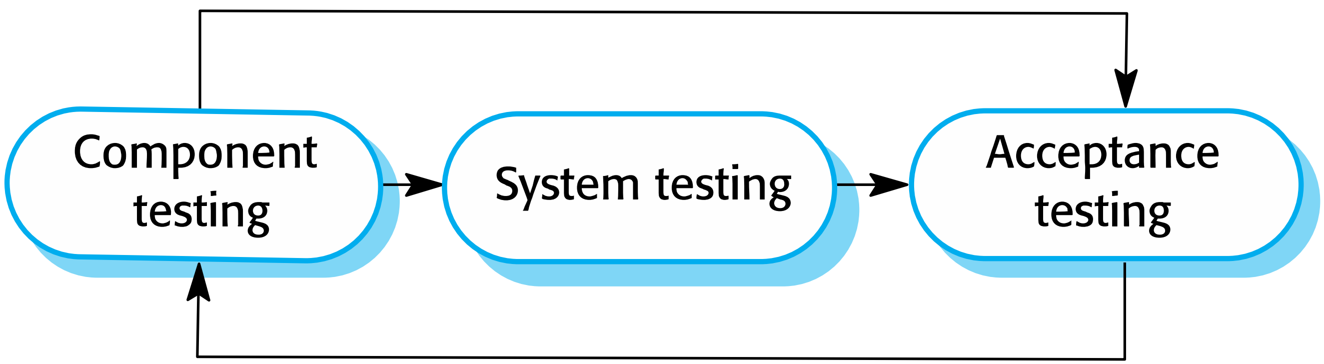
System implementation

- ✧ The software is implemented either by developing a program or programs or by configuring an application system.
- ✧ Design and implementation are interleaved activities for most types of software system.
- ✧ Programming is an individual activity with no standard process.
- ✧ Debugging is the activity of finding program faults and correcting these faults.

Software Verification & validation

- Verification and validation (V & V) is intended to show that a system conforms to its specification and meets the requirements of the system customer.
- Verification refers to the set of activities that ensure software correctly implements the specific function. Validation refers to the set of activities that ensure that the software that has been built is traceable to customer requirements
- Involves checking and review processes and system testing.
- System testing involves executing the system with test cases that are derived from the specification of the real data to be processed by the system.
- Testing is the most commonly used V & V activity.

Stages of testing

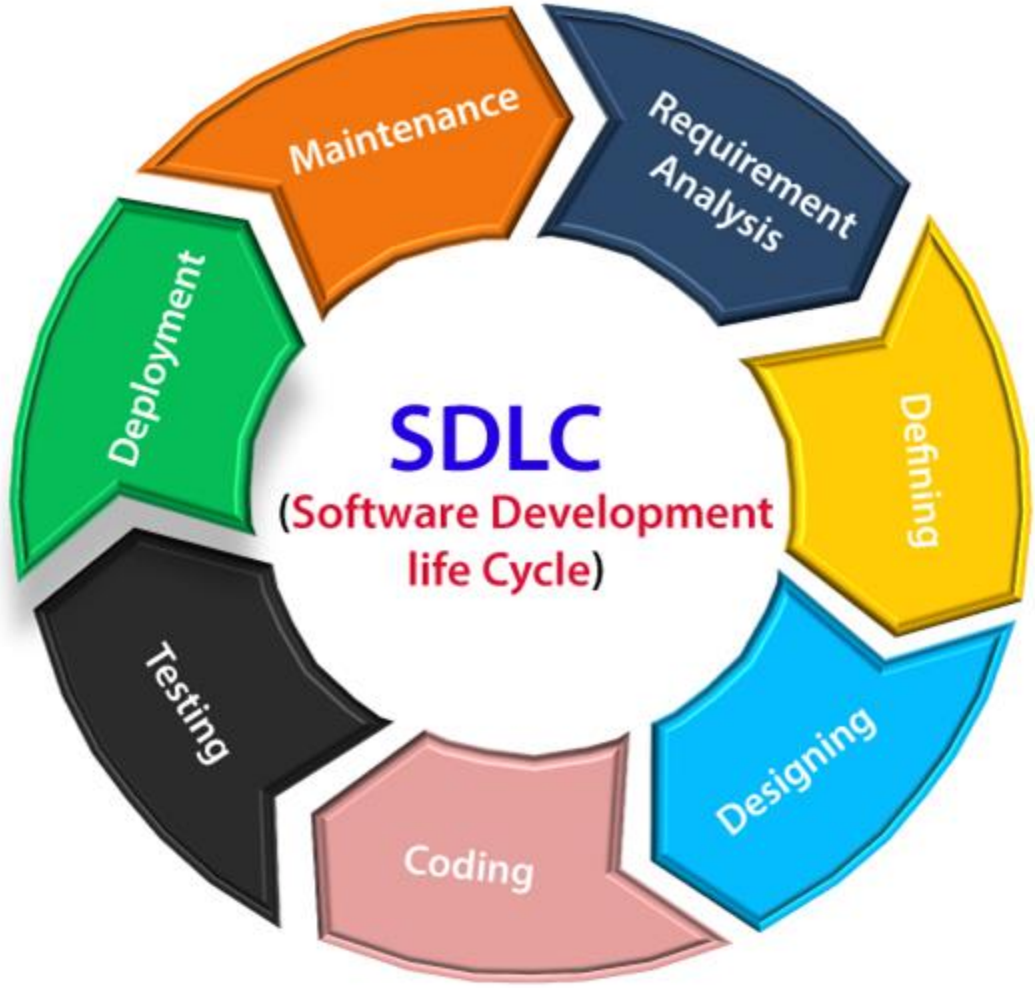


Software process models
Software Development Life Cycle (SDLC)

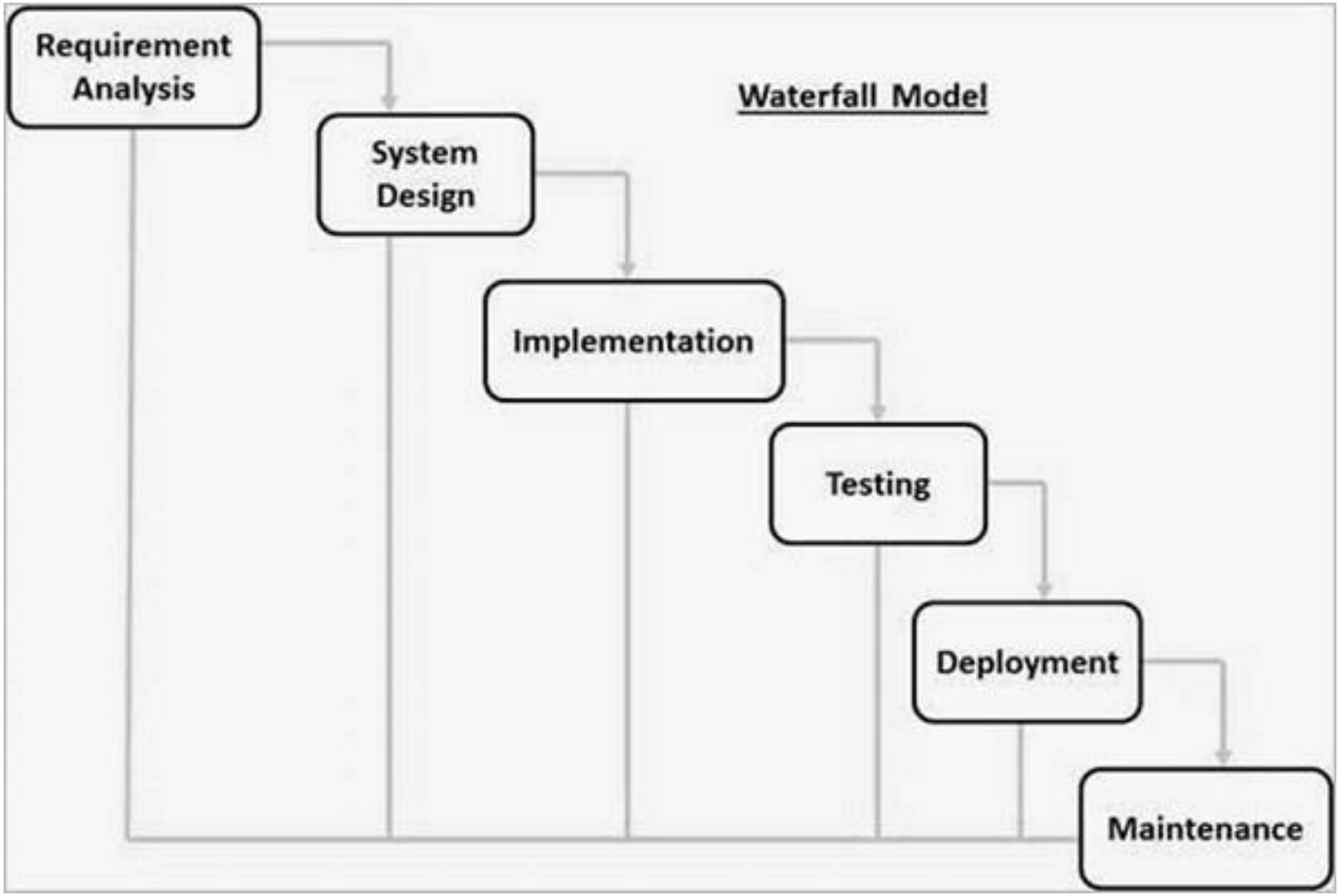
Software Development Life Cycle (SDLC)

- A software life cycle model (also termed process model) is a pictorial and diagrammatic representation of the software life cycle.
- A life cycle model represents all the methods required to make a software product transit through its life cycle stages.
- It also captures the structure in which these methods are to be undertaken

SDLC Cycle


Maintenance
Requirement
Analysis
Defining
Designing
Coding
Testing
Deployment
SDLC
(Software Development
life Cycle)

The waterfall model



Waterfall model phases

- There are separate identified phases in the waterfall model:
  - Requirements analysis and definition
  - System and software design
  - Implementation and unit testing
  - Integration and system testing
  - Operation and maintenance
- The main drawback of the waterfall model is the difficulty of accommodating change after the process is underway.
- In principle, a phase has to be complete before moving onto the next phase.

The waterfall model

✧ Advantages

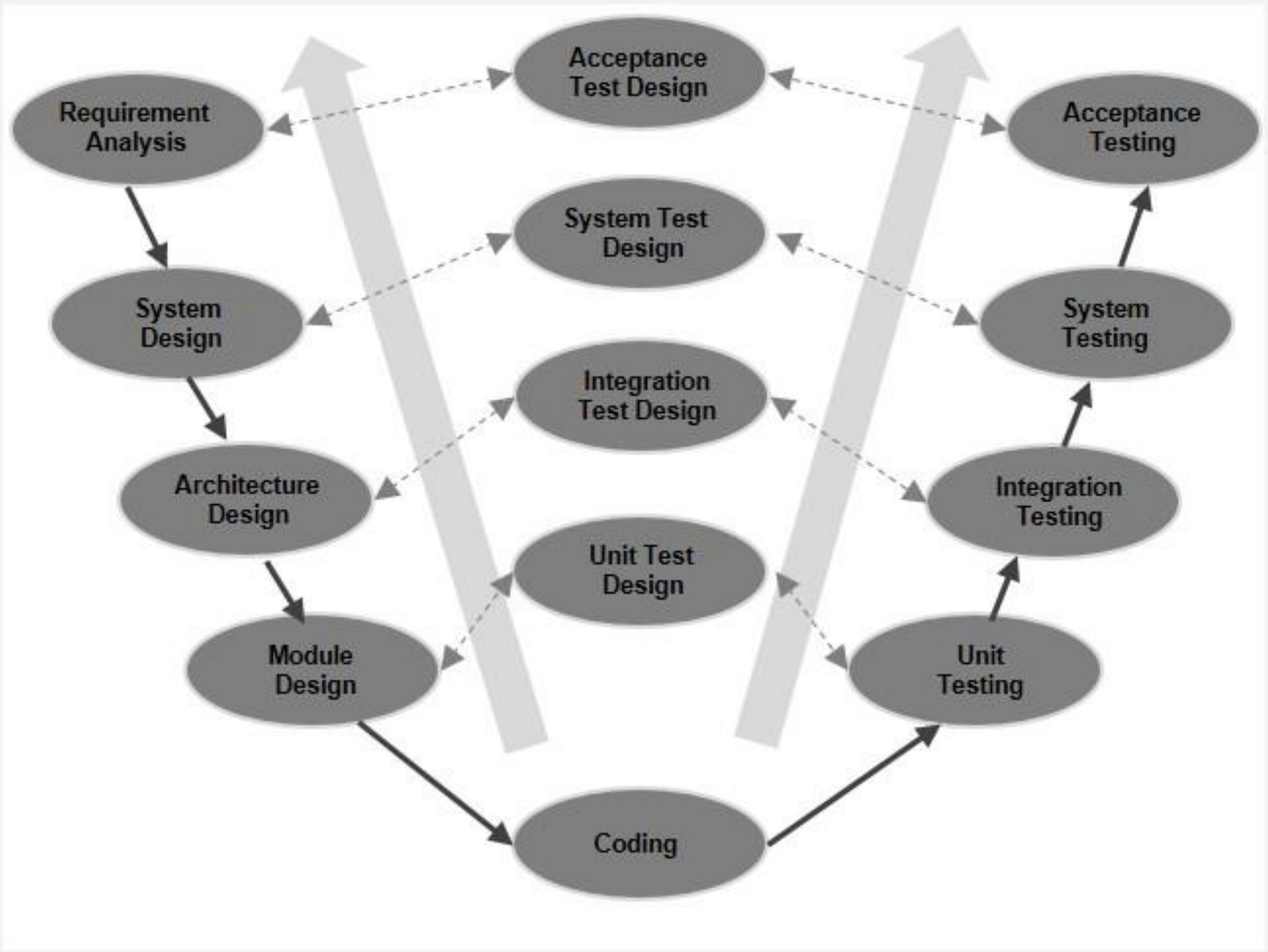
- Simple and easy to understand and use
- Easy to manage due to the rigidity of the model. Each phase has specific deliverables and a review process.
- Phases are processed and completed one at a time.
- Works well for smaller projects where requirements are very well understood.
- Clearly defined stages.
- Well understood milestones.
- Easy to arrange tasks.
- Process and results are well documented.

The waterfall model

✧ Disadvantages

- No working software is produced until late during the life cycle.
- High amounts of risk and uncertainty.
- Not a good model for complex and object-oriented projects.
- Poor model for long and ongoing projects.
- Cannot accommodate changing requirements.

V-Model


Requirement Analysis
System Design
Architecture Design
Module Design
Coding
Acceptance Test Design
System Test Design
Integration Test Design
Unit Test Design
Acceptance Testing
System Testing
Integration Testing
Unit Testing

V-Model

- The V-model is a type of model where process executes in a sequential manner in V-shape.
- It is also known as Verification and Validation model.
- It is based on the association of a testing phase for each corresponding development stage.
- Development of each step directly associated with the testing phase.
- The next phase starts only after completion of the previous phase i.e. for each development activity, there is a testing activity corresponding to it.

V-Model

✧ Advantages

- This is a highly disciplined model and Phases are completed one at a time.
- V-Model is used for small projects where project requirements are clear.
- Simple and easy to understand and use.
- This model focuses on verification and validation activities early in the life cycle thereby enhancing the probability of building an error-free and good quality product.
- It enables project management to track progress accurately.

V-Model

✧ Disadvantages

- High risk and uncertainty.
- It is not a good for complex and object-oriented projects.
- It is not suitable for projects where requirements are not clear and contains high risk of changing.
- This model does not support iteration of phases.
- It does not easily handle concurrent events.

Iterative process model

- ✧ In this Model, you can start with some of the software specifications and develop the first version of the software.
- ✧ After the first version if there is a need to change the software, then a new version of the software is created with a new iteration.
- ✧ Every release of the Iterative Model finishes in an exact and fixed period that is called iteration.

Iterative process model

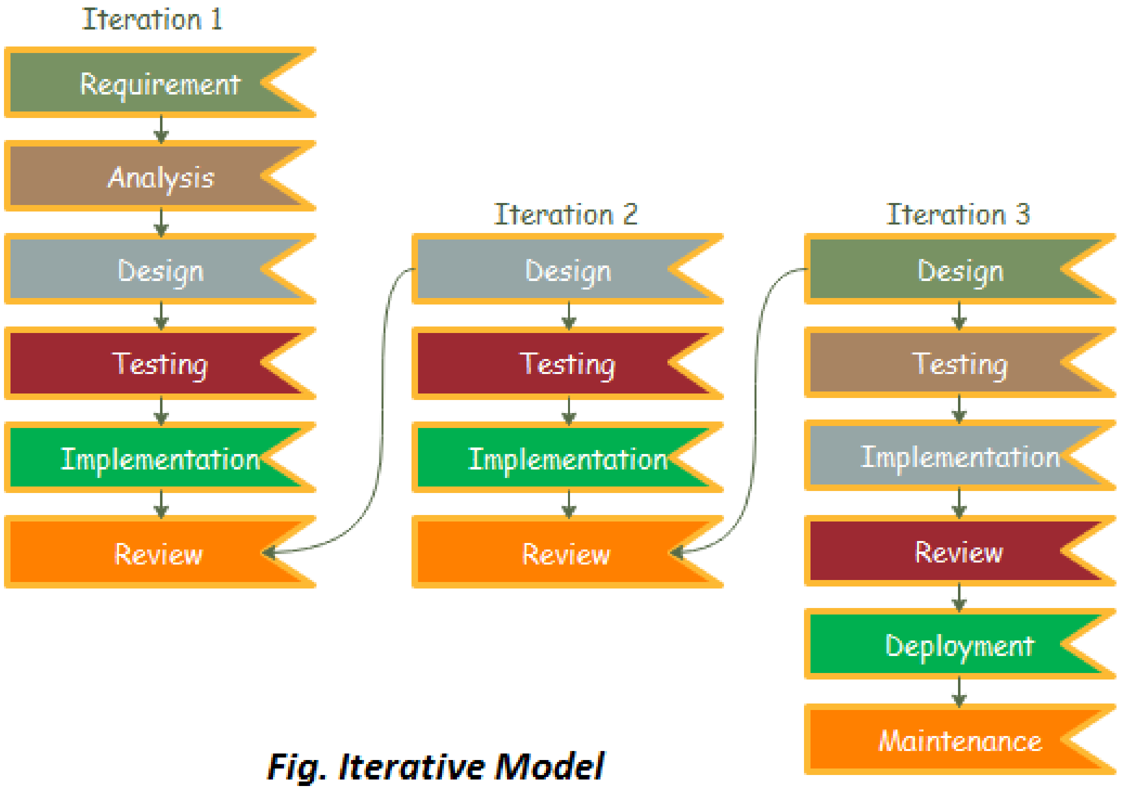


Fig. Iterative Model

Iterative process model

✧ Advantages

- Some working functionality can be developed quickly and early in the life cycle.
- Results are obtained early and periodically.
- Parallel development can be planned.
- Progress can be measured.
- Less costly to change the scope/requirements.
- Testing and debugging during smaller iteration is easy.

Iterative process model

- Disadvantages
  - More resources may be required.
  - Although cost of change is lesser, but it is not very suitable for changing requirements.
  - More management attention is required.
  - Defining increments may require definition of the complete system.
  - Not suitable for smaller projects.

Incremental process model

- ✧ Incremental Model is a process of software development where requirements divided into multiple standalone modules of the software development cycle.
- ✧ In this model, each module goes through the requirements, design, implementation and testing phases.
- ✧ Every subsequent release of the module adds function to the previous release.
- ✧ The process continues until the complete system achieved.

Incremental process model

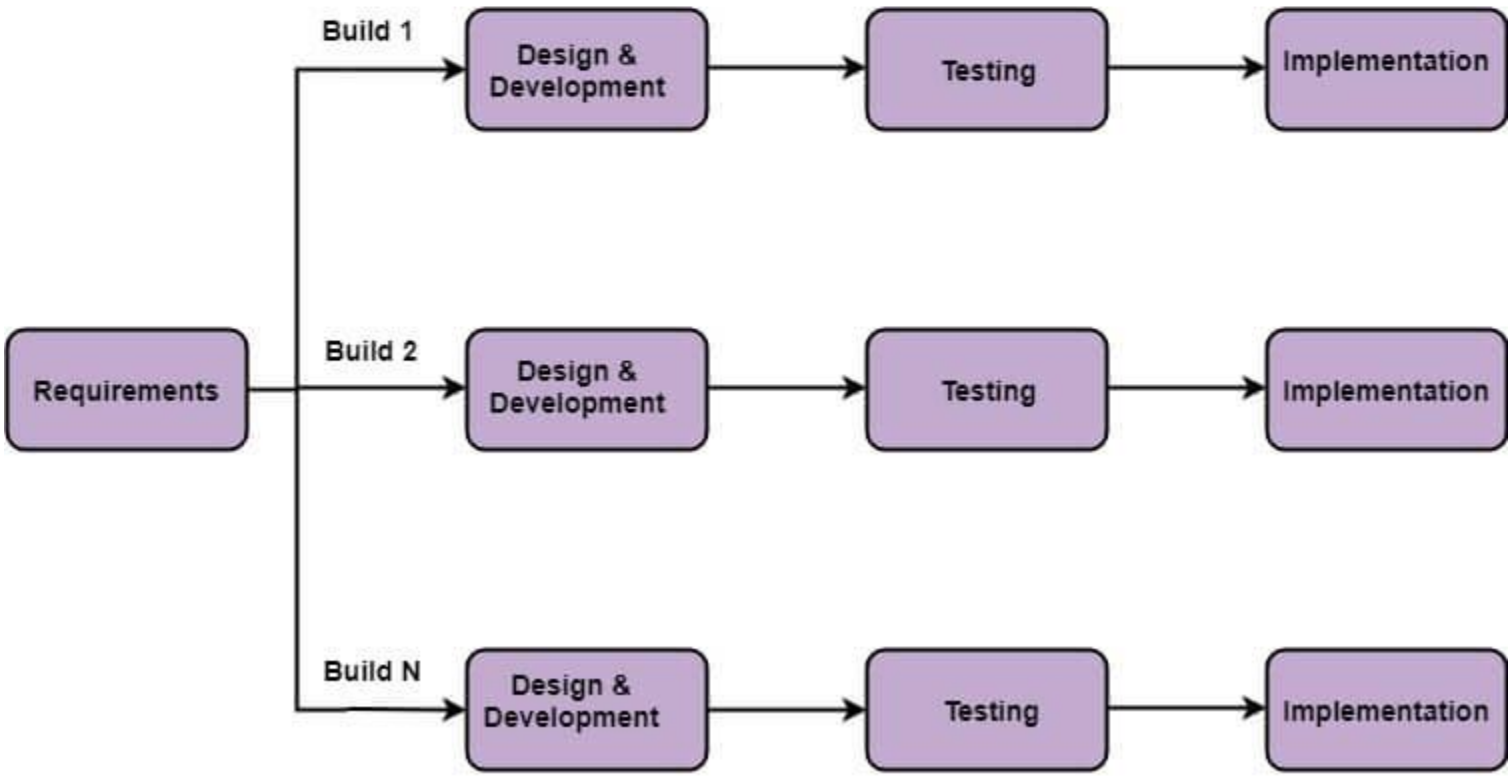


Fig: Incremental Model

Incremental development

✧ Advantage

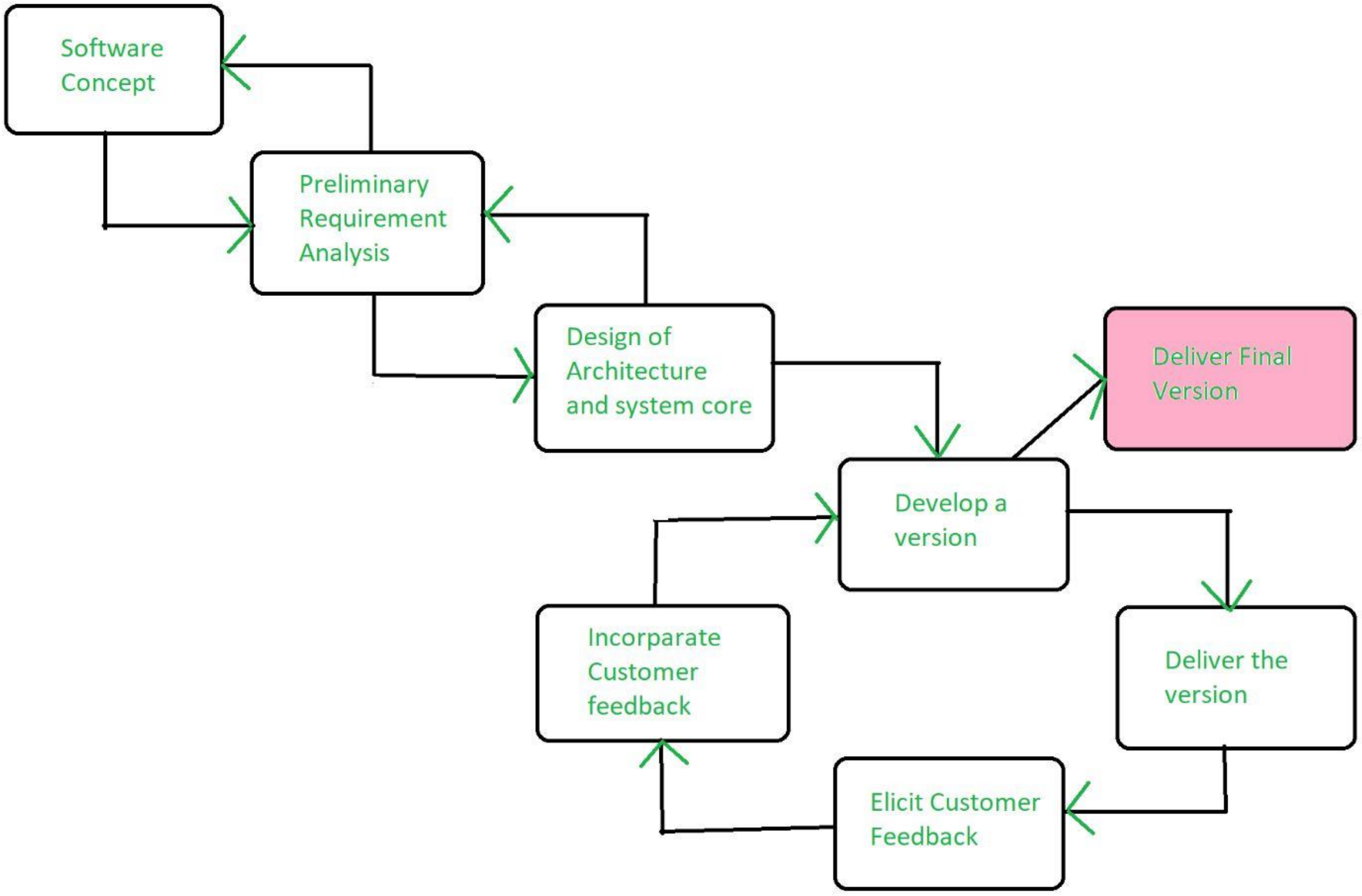
- Errors are easy to be recognized.
- Easier to test and debug
- More flexible.
- Simple to manage risk because it handled during its iteration.
- The Client gets important functionality early.

Incremental development

- ✧ Disadvantage
  - Need for good planning
  - Total Cost is high.
  - Well defined module interfaces are needed.

Evolutionary process models

Evolutionary process



Evolutionary process

- ✧ Evolutionary model is a combination of Iterative and Incremental model of software development life cycle.
- ✧ Delivering your system in a big bang release, delivering it in incremental process over time is the action done in this model.
- ✧ Some initial requirements and architecture envisioning need to be done.
- ✧ It is better for software products that have their feature sets redefined during development because of user feedback and other factors.
- ✧ The Evolutionary development model divides the development cycle into smaller, incremental waterfall models in which users are able to get access to the product at the end of each cycle

Evolutionary process

- Advantages
  - In evolutionary model, a user gets a chance to experiment partially developed system.
  - It reduces the error because the core modules get tested thoroughly.
- Disadvantages
  - Sometimes it is hard to divide the problem into several versions that would be acceptable to the customer which can be incrementally implemented and delivered.

The Spiral Model

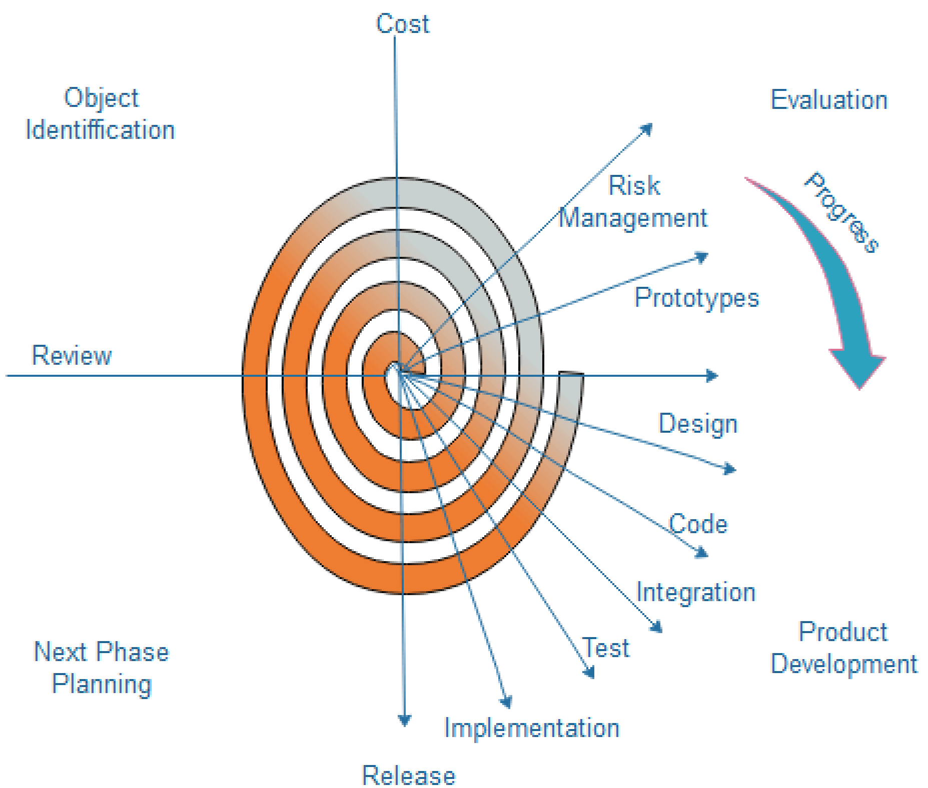
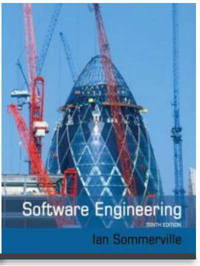


Fig. Spiral Model

The Spiral Model



- The spiral model, initially proposed by Boehm, is an evolutionary software process model that couples the iterative feature of prototyping with the controlled and systematic aspects of the linear sequential model.
- It implements the potential for rapid development of new versions of the software.
- Using the spiral model, the software is developed in a series of incremental releases.
- During the early iterations, the additional release may be a paper model or prototype.
- During later iterations, more and more complete versions of the engineered system are produced.

The Spiral Model

✧ Advantages

- High amount of risk analysis
- Useful for large and mission-critical projects.

✧ Disadvantages

- Can be a costly model to use.
- Risk analysis needed highly particular expertise
- Doesn't work well for smaller projects